

What happens when coupling switches on?

exp085 · 17 August 2026 · Draft

Abstract

Two slightly detuned PING circuits begin from the same mature uncoupled trajectory before reciprocal excitation switches on. Across 11 coupling strengths, the unwrapped relative phase changes from a persistent drift of 30.1 rad/s to -0.01 rad/s. The experiment uses runtime-state branching to isolate coupling onset from network activation and shows the transition directly without imposing a binary synchronization threshold.

Methods

1. **Equilibrate two detuned circuits.** Independent 100 Hz/channel Poisson streams drive two 80 E / 20 I PING components. Circuit A uses 4 ms inhibitory decay and circuit B uses 5 ms. Both run uncoupled for 1000 ms.
2. **Branch one mature state.** The executor captures voltages, refractory counters, conductances, population histories, and input-delay histories at coupling onset. That identical state branches across 11 reciprocal coupling strengths. Private future inputs remain paired across K and independent between circuits.
3. **Measure relative phase.** E-population rates are smoothed with a 5 ms Gaussian kernel and filtered over 20–60 Hz. Hilbert phases define the unwrapped A-minus-B relative phase, expressed as its change from coupling onset. The final 100 ms are excluded to prevent the finite-record filter boundary from appearing as a phase jump. This continuous trajectory is the primary result; no locking threshold or latency is inferred.
4. **Measure supporting responses.** The registered `exp085-dominant-rhythm-v1` policy reports each circuit’s dominant rhythm from the final 1.5 s. Frequency difference, descriptive phase-locking value, and population firing rates support the phase trajectories.

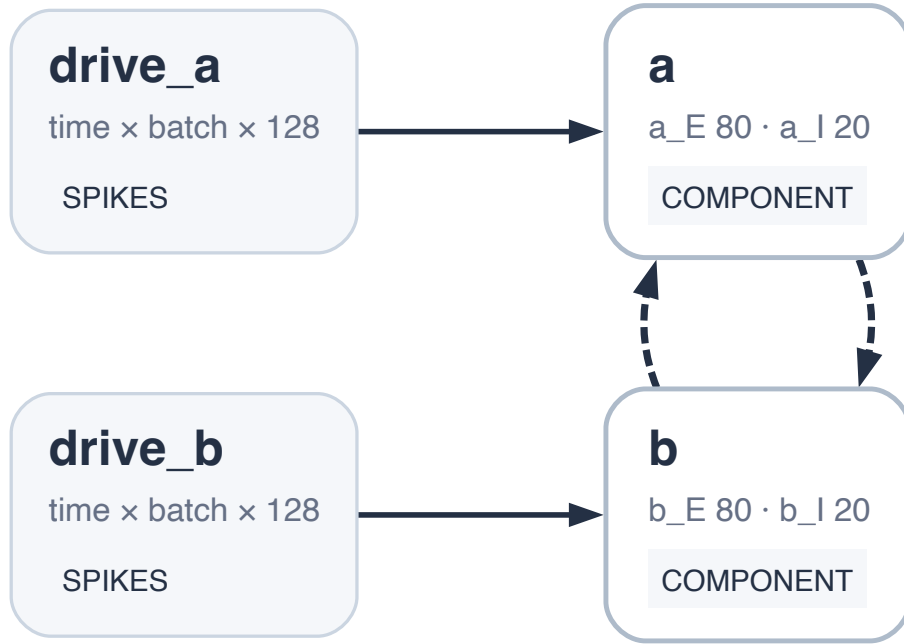


Figure 1: Two-circuit coupling-onset graph. Independent spike inputs drive circuits A and B. Four reciprocal AMPA projections share coupling strength K ; their weights are zero during equilibration and vary only after the common runtime-state branch. Local PING topology and every non-coupling parameter remain fixed.

Results

The relative-phase trajectories expose the response to coupling without reducing it to a label. At $K = 0$ the circuits repeatedly lap one another. Increasing K changes the slope and structure of that drift; the strongest condition ends with a median phase-locking value of 0.99.

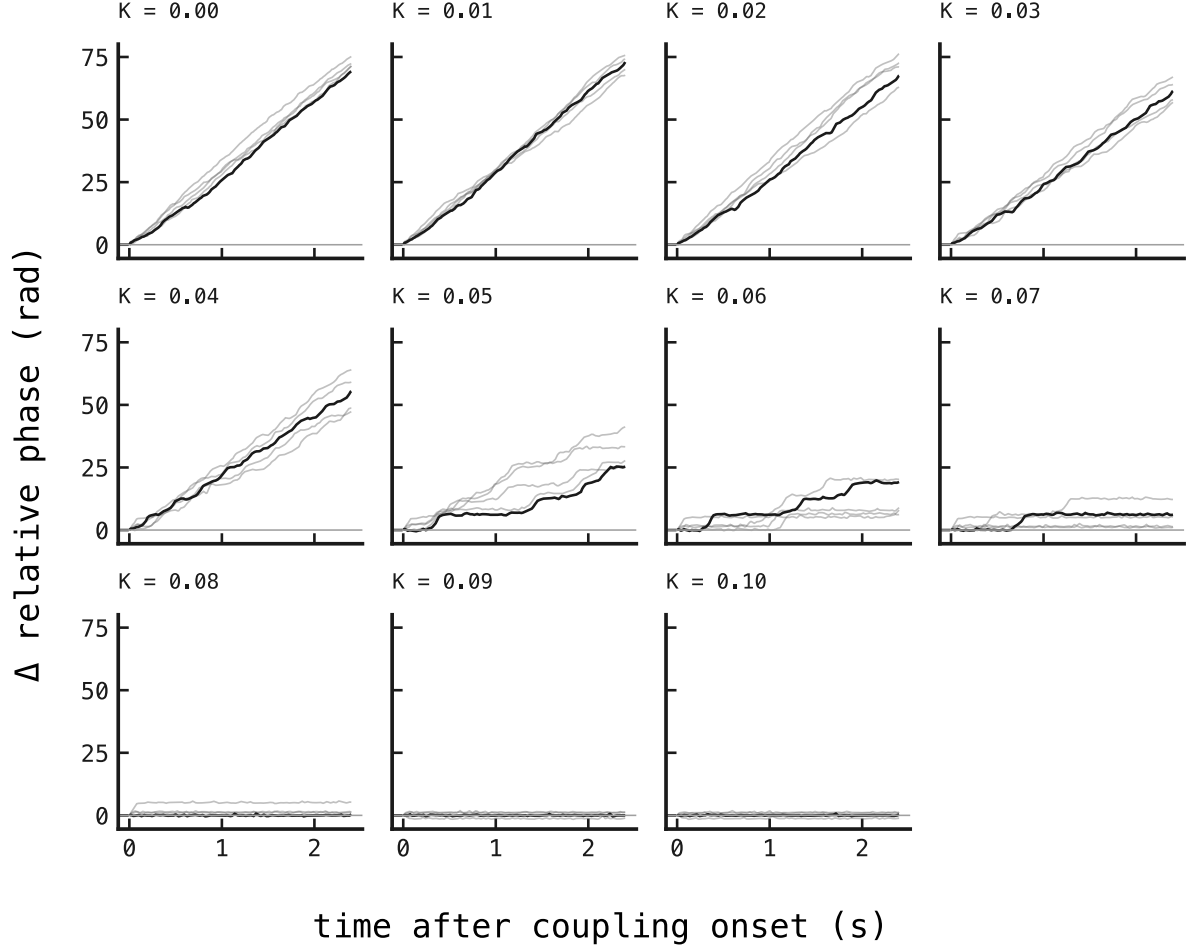


Figure 2: Change in unwrapped A-minus-B relative phase after coupling onset. Time is in seconds and phase change is in radians. Panels increase in reciprocal coupling K from left to right and top to bottom. Every trace is zeroed at onset; the final 100 ms are excluded as a registered filter-edge guard. The black trace is one predeclared paired trial; grey traces are the other 4 trials from the same mature-state and input protocol. Persistent slope indicates phase drift, plateaus indicate transient capture, and a bounded trace indicates a stable phase relationship.

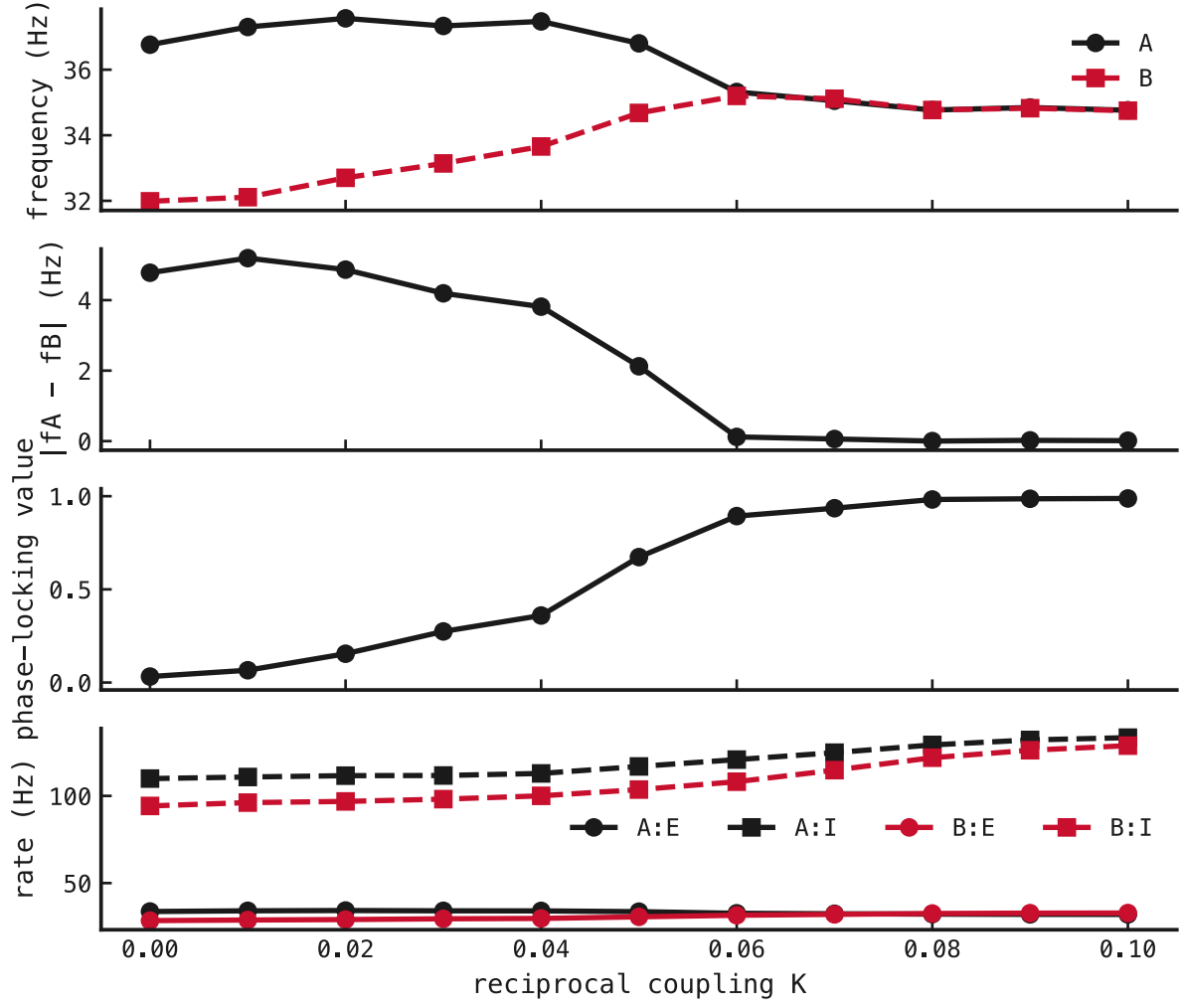


Figure 3: Supporting coupling response. First: median dominant E-population frequencies of circuits A and B in hertz. Second: their absolute difference in hertz. Third: median descriptive phase-locking value across 5 paired trials. Fourth: mean E and I firing rates in hertz for both circuits. K is the shared weight of the four reciprocal AMPA projections.

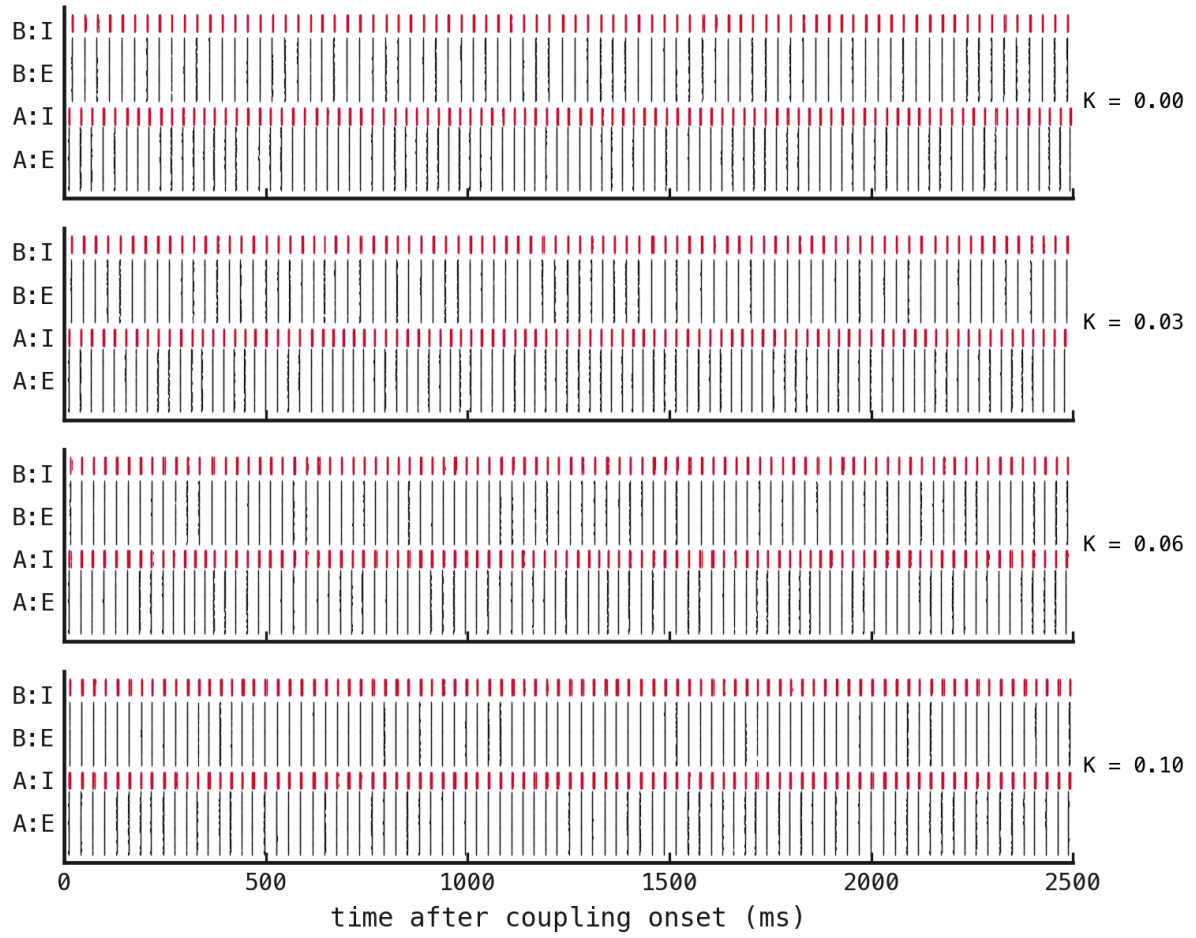


Figure 4: Population rasters after coupling onset at the four predeclared K values. Time is in milliseconds; black marks identify E cells and red marks identify I cells. Each panel stacks A:E, A:I, B:E, and B:I from bottom to top for the same paired trial. The sequence shows how population volleys change as reciprocal excitation strengthens.